Reinforcement Learning-Based Personalized Depression Treatment Using Synthetic Data and Real-Time Decision Support

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CLINICAL & AGENTIC AI TAKEAWAY

Reinforcement Learning-based personalized depression treatment using synthetic data and real-time decision support can optimize antidepressant treatment strategies, reducing side effects and healthcare costs.

ABSTRACT

Depression treatment often involves a complex and lengthy trial-and-error process, where clinicians sequentially prescribe medications to identify the most effective treatment for each patient.This approach can lead to delayed recovery, unnecessary side effects, and increased healthcare costs.To address this challenge, we present a Reinforcement Learning (RL)-based framework designed to optimize antidepressant treatment strategies through dynamic, patient-specific decision-making.The proposed system leverages synthetically generated patient data to simulate real-world treatment scenarios while ensuring privacy and scalability.A synthetic depression patient simulator was developed to model daily symptom trajectories influenced by medication type, dosage, adherence, side effects, and stochastic life events.This simulated data allowed the training of a deep Q-learning agent within a custom-built reinforcement learning environment.The agent learned to recommend treatment adjustments or continuations based on temporal symptom patterns and treatment history.Key components of the framework include experience replay, target model updates, and an epsilon-greedy exploration strategy to balance exploration and exploitation during training.The system was evaluated using unseen synthetic patients to assess generalization performance.Comprehensive visual analyses were conducted to characterize the symptom distribution, medication assignment, agent reward dynamics, and real-time treatment recommendations.The real-time recommendation system demonstrated the ability to provide timely, personalized treatment suggestions, switching medications when appropriate and maintaining stability when patient symptoms improved.The model's decision-making process is closely aligned with clinical reasoning, supporting its potential as a decision support tool in precision psychiatry.This study offers a privacypreserving, scalable, and clinically relevant pathway for optimizing depression How to cite this paper: de

Study Overview & Background

Depression treatment often involves a complex and lengthy trial-and-error process, where clinicians sequentially prescribe medications to identify the most effective treatment for each patient.This approach can lead to delayed recovery, unnecessary side effects, and increased healthcare costs.To address this challenge, we present a Reinforcement Learning (RL)-based framework designed to optimize antidepressant treatment strategies through dynamic, patient-specific decision-making.The proposed system leverages synthetically generated patient data to simulate real-world treatment

scenarios while ensuring privacy and scalability.A synthetic depression patient simulator was developed to model daily symptom trajectories influenced by medication type, dosage, adherence, side effects, and stochastic life events.This simulated data allowed the training of a deep Q-learning agent within a custom-built reinforcement learning environment.The agent learned to recommend treatment adjustments or continuations based on temporal symptom patterns and treatment history.Key components of the framework include experience replay, target model updates, and an epsilon-greedy exploration strategy to balance exploration and exploitation during training.The system was evaluated using unseen synthetic patients to assess generalization performance.Comprehensive visual analyses were conducted to characterize the symptom distribution, medication assignment, agent reward dynamics, and real-time treatment recommendations.The real-time recommendation system demonstrated the ability to provide timely, personalized treatment suggestions, switching medications when appropriate and maintaining stability when patient symptoms improved.The model's decision-making process is closely aligned with clinical reasoning, supporting its potential as a decision support tool in precision psychiatry.This study offers a privacypreserving, scalable, and clinically relevant pathway for optimizing depression How to cite this paper: de

Methodology & Clinical Focus

Investigation into artificial intelligence models, machine learning algorithms, and clinical predictive tools applied to orthopedic surgery and musculoskeletal medicine.

Key Findings & Analysis

Reinforcement Learning-based personalized depression treatment using synthetic data and real-time decision support can optimize antidepressant treatment strategies, reducing side effects and healthcare costs.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

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